IMAGS

Music-Induced Analgesia Genome Study

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Credit: Verywell / Lara Antal

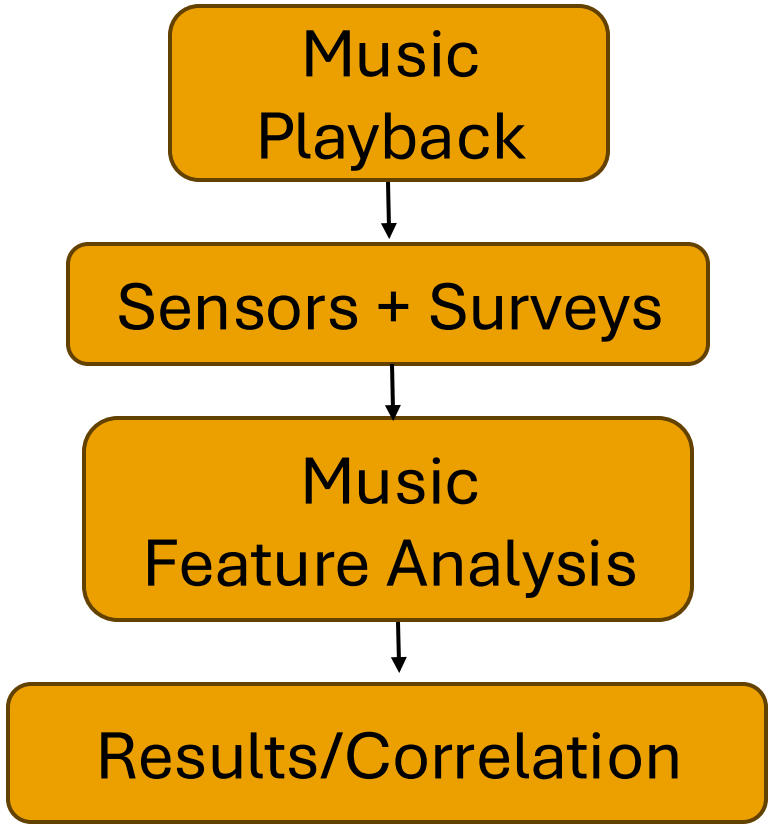
IMAGS: Music-Induced Analgesia Genome Study

- How does music reduce chronic pain?
- Combines music listening, physiological data, and self-reported pain surveys

More Than Just Entertainment by M. Murray, MM, MT-BC, June 29, 2022, *NAMI*

Project Goals: Expanding IMAGS

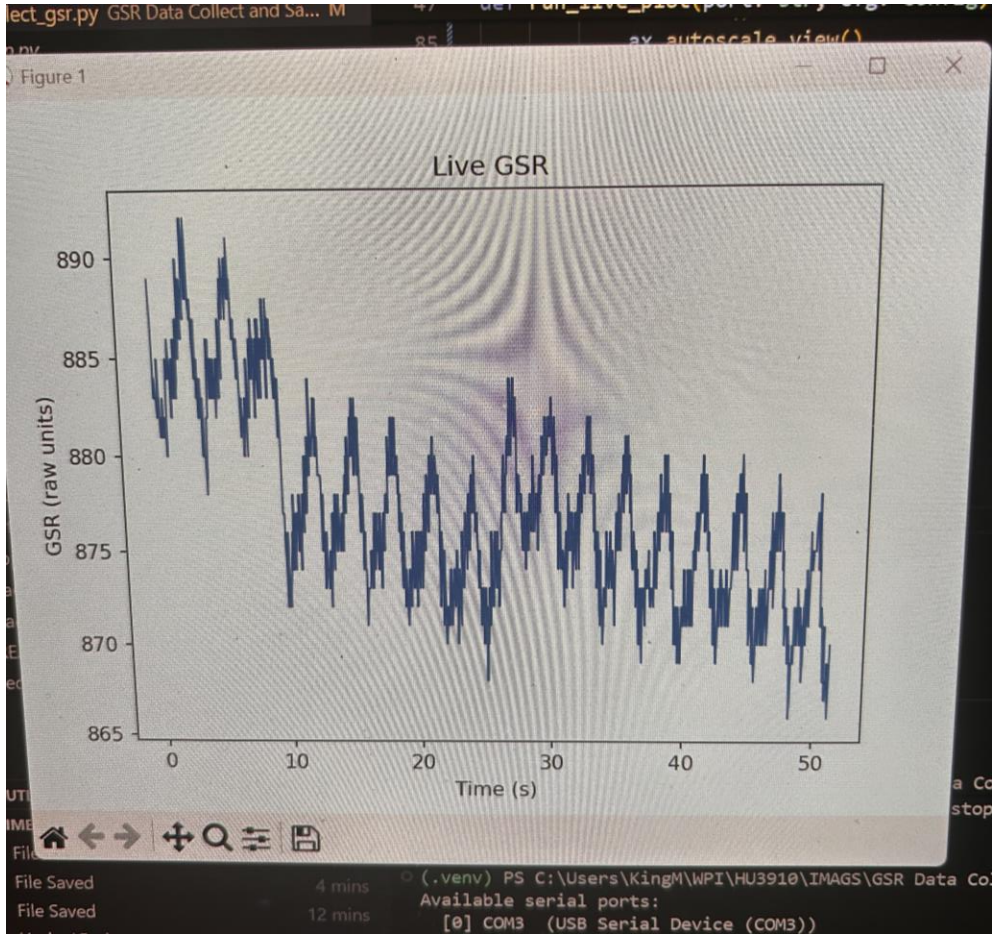
- Analyze musical features at moments of reduced pain awareness
- Explore different pain indicators/sensors
- Develop a music interface that shows:
  - Physiological Responses
  - Pain Surveys
  - Extracted Musical Features



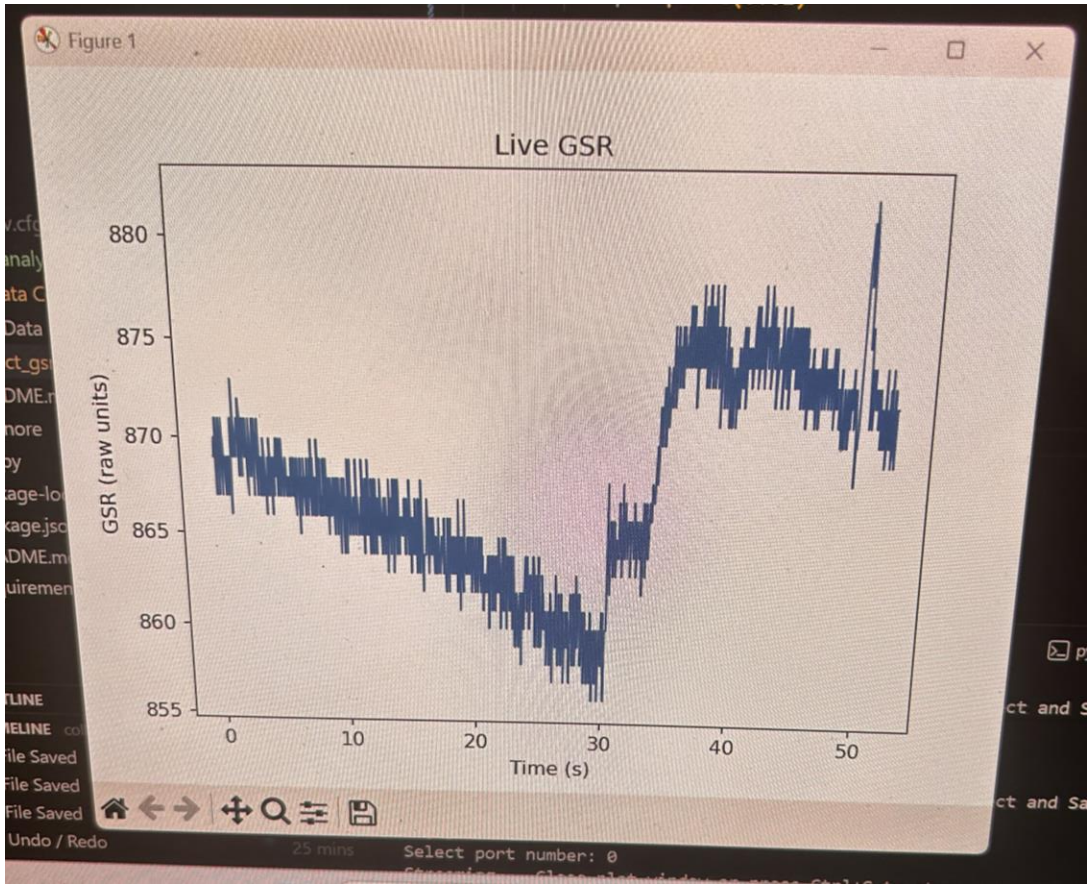
Problems and Adapting

- Starting from scratch
  - New GSR
  - New Software
  - No prior research
- Environmental noise in GSR data

Environmental Noise



Without Charger



With Charger

Member Concentrations

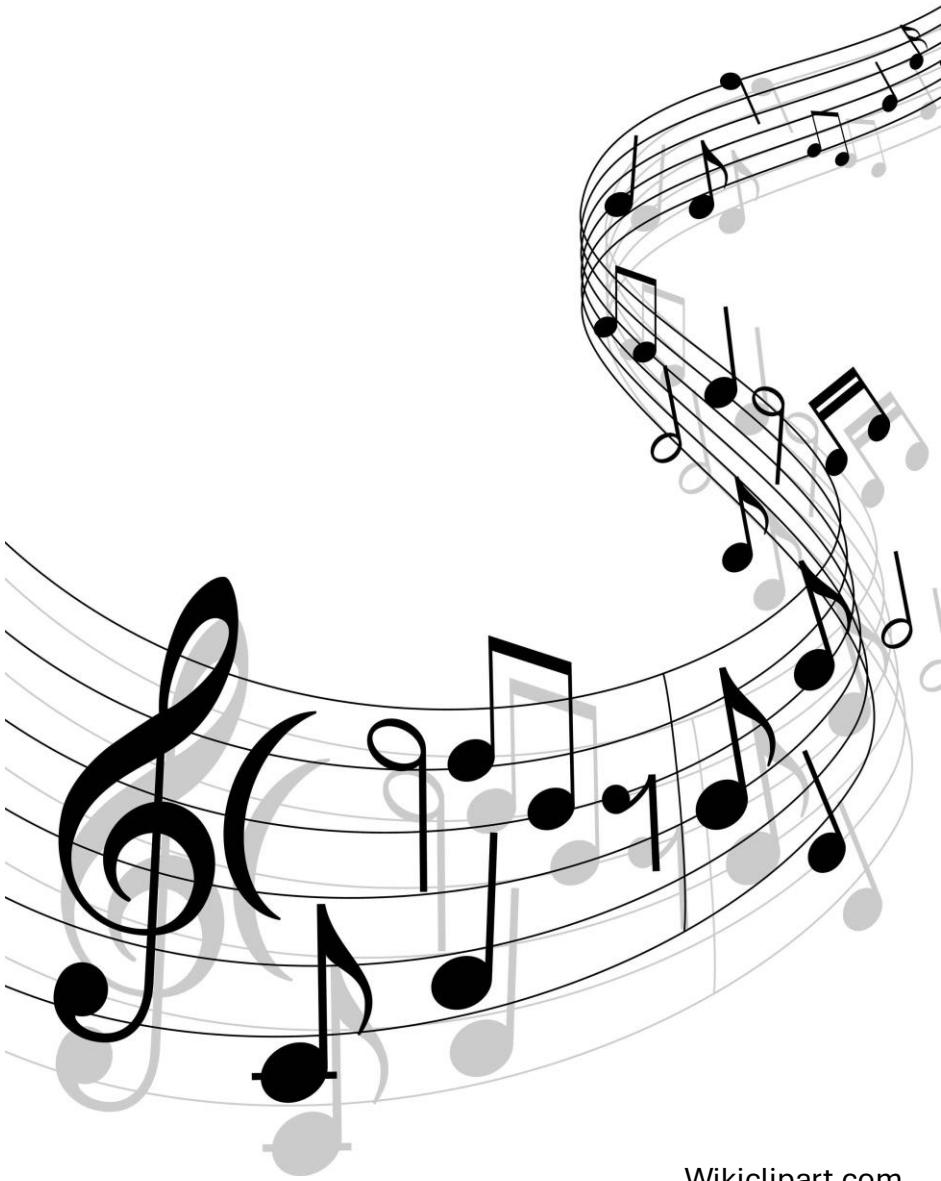
- **Dylan and Jadon**
  - Software
- **Charlie**
  - Research and Presentations
- **Sean**
  - Hardware

Music Information Retrieval (MIR)

- Used to determine song characteristics
- Can determine items such as:
  - Key
  - Tempo
  - Volume
  - Mood/Genre
- Our open-source software
  - Essentia

Parts of Music to Measure

- Familiarity
- Global Tempo/BPM
- Mood
- Dynamic Range
- Loudness
- Energy



Wikiclipart.com

Survey Questions

- Prior to testing:
  - Rating pain levels
  - Familiarity and experience with music
- After testing:
  - Rating pain levels and changes
  - Explaining experiences and engagement
- GSR:
  - Environment related such as room temp or time of day
  - Patient related such as last meal or caffeine use

Measuring Chronic Pain

- Measuring pain is difficult, especially in chronic pain patients.
- Chronic pain is intertwined with stress and can sometimes be exacerbated by stress responses.[1]
- Both self-reported pain/stress data and physiological indicators are often used to evaluate pain.[1]
- To get an accurate measure of perceived pain, we should collate our sensor and user-reported data prior to processing.

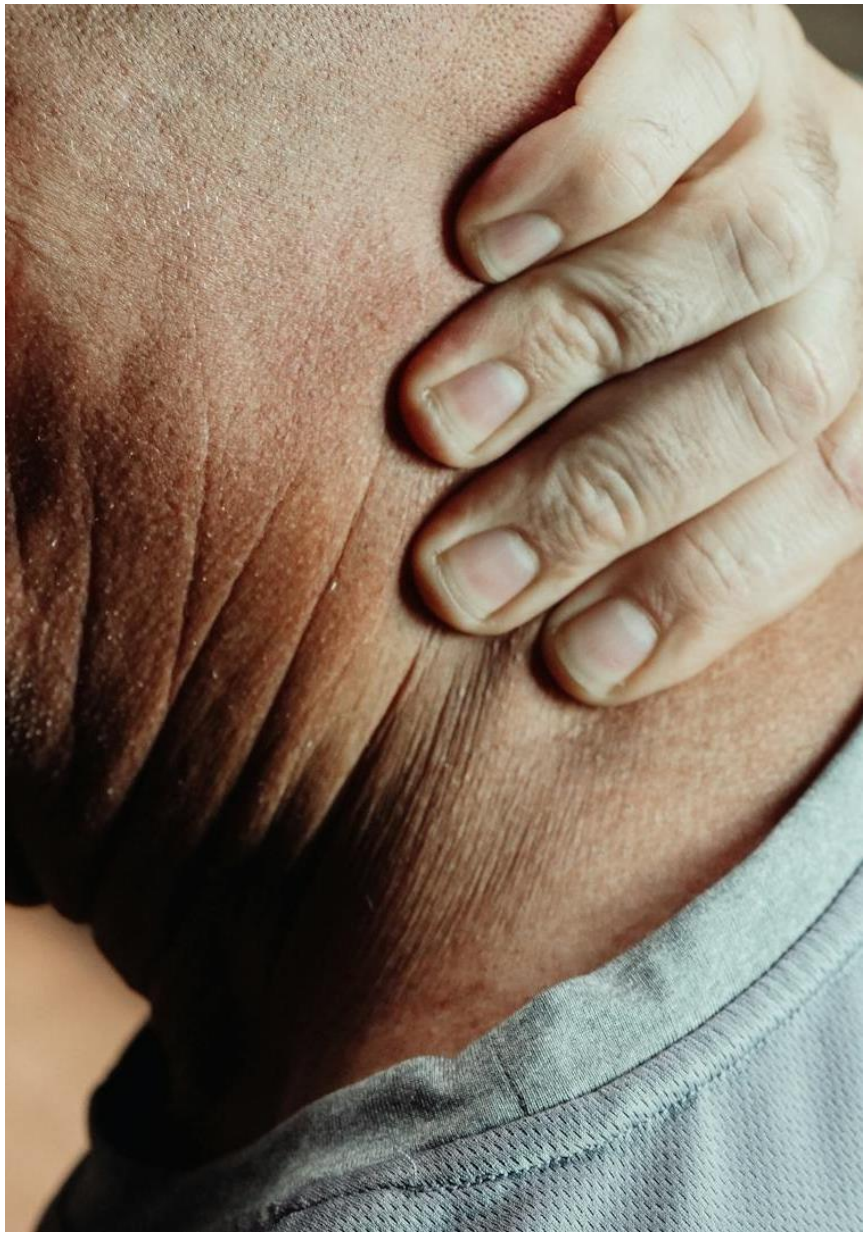
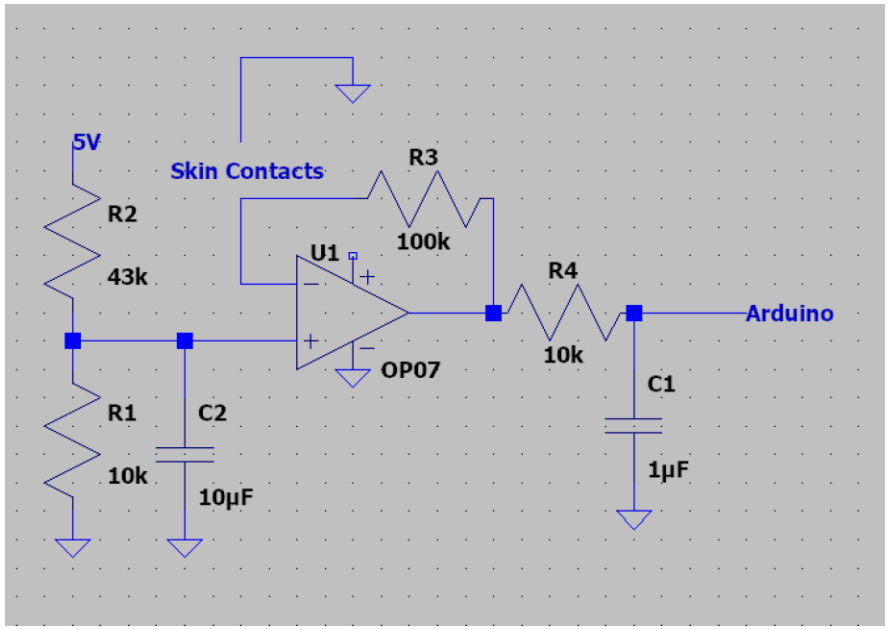
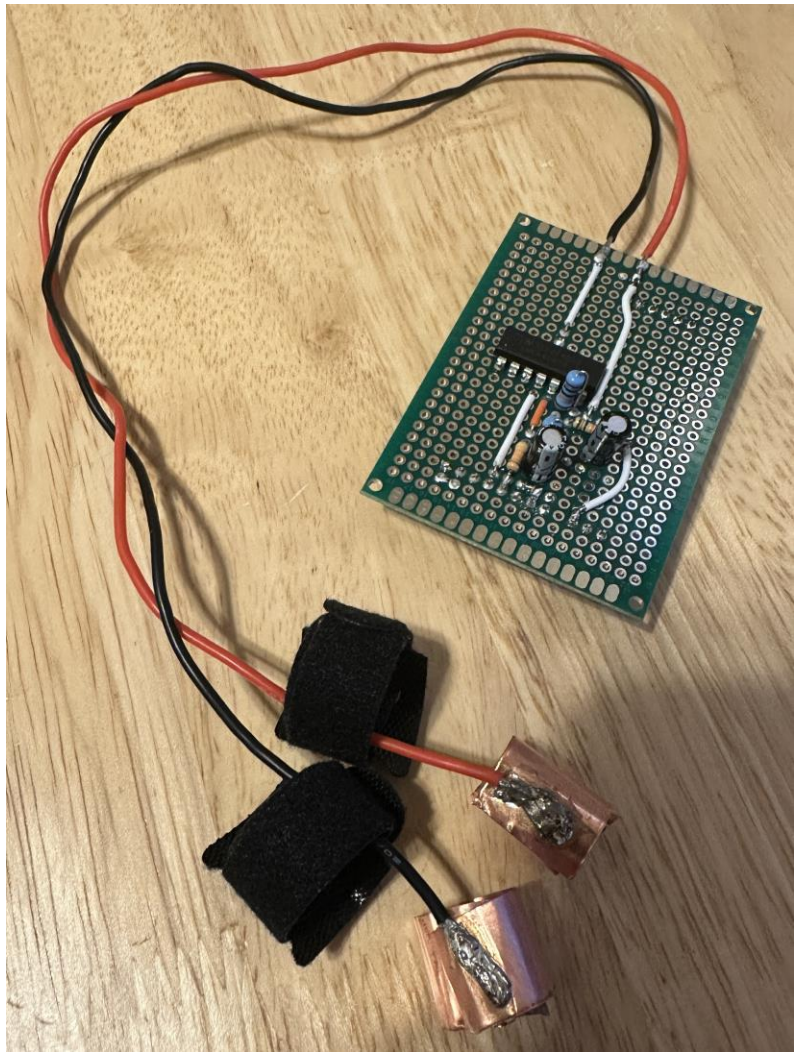
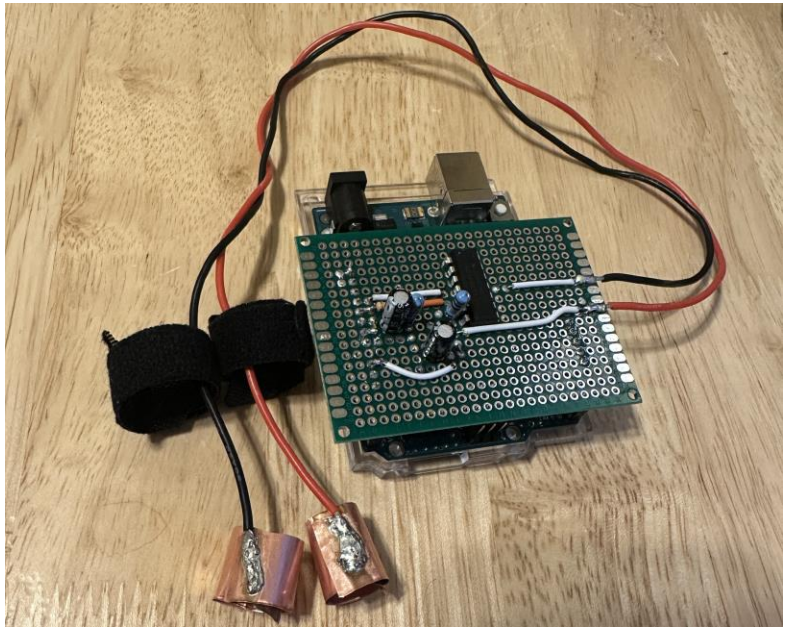


Photo by Kindel Media (2)


5V
Skin Contacts
R3
100k
R2
43k
U1
R4
10k
Arduino
OP07
R1
10k
C2
10µF
C1
1µF

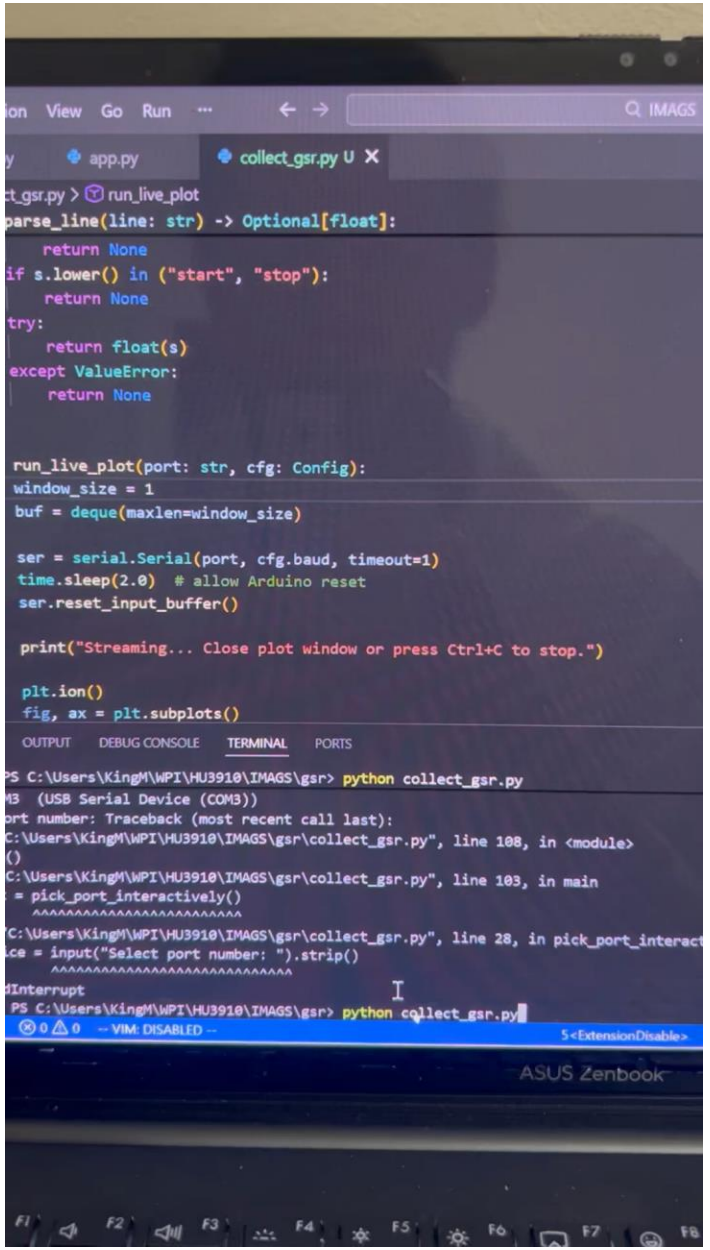




Our GSR

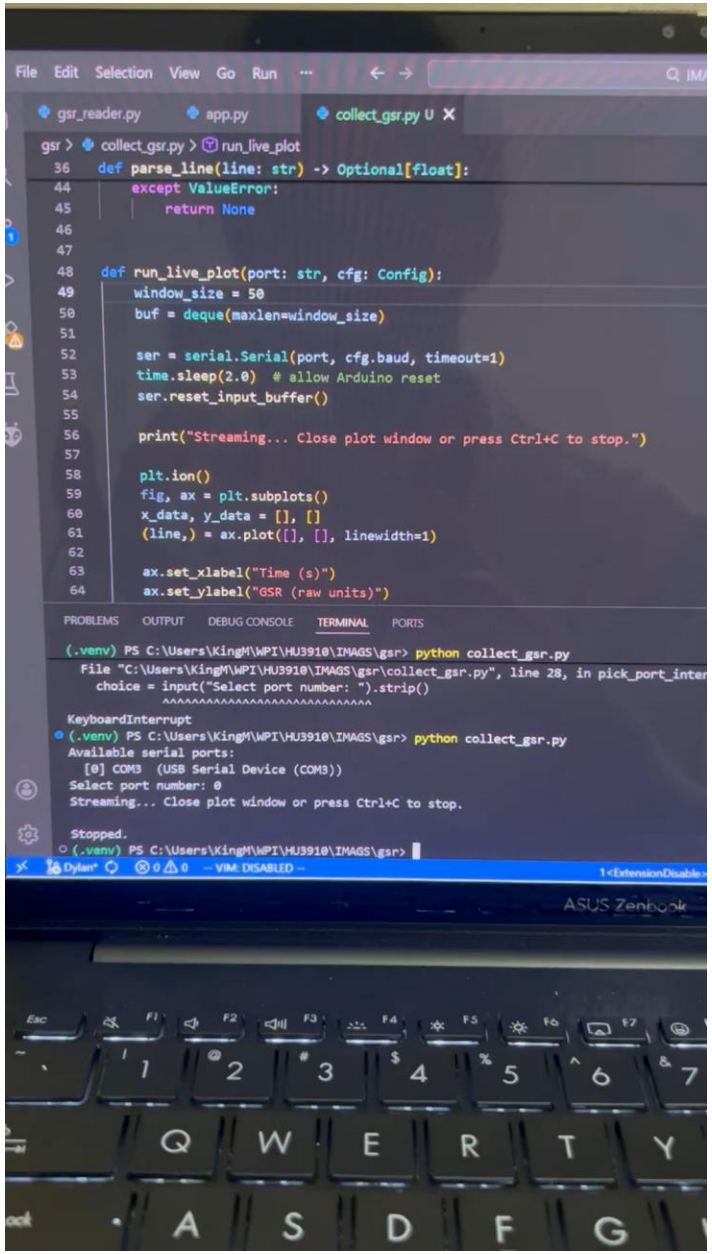
Live GSR plotting:

Raw GSR

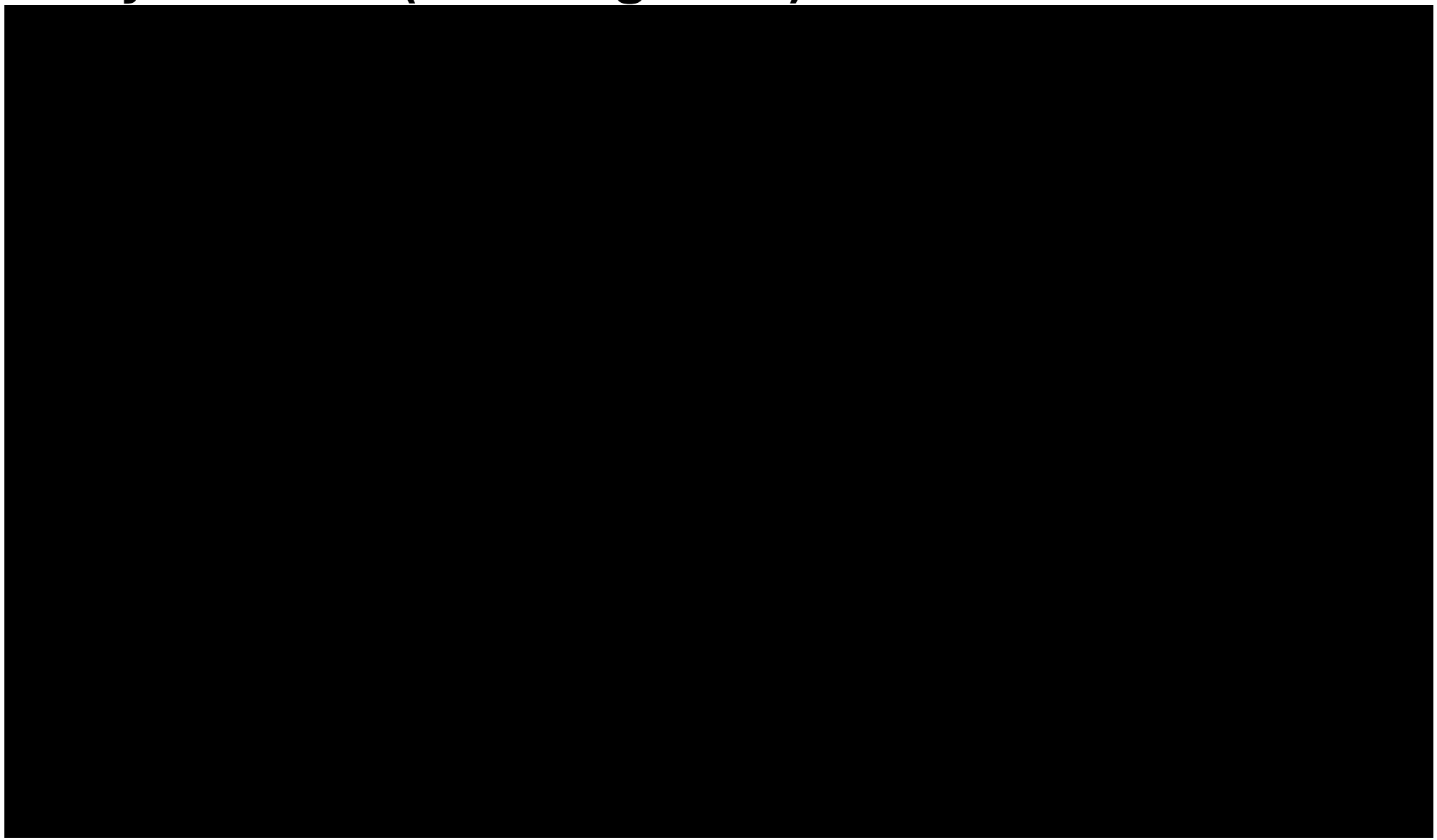


Live GSR plotting:

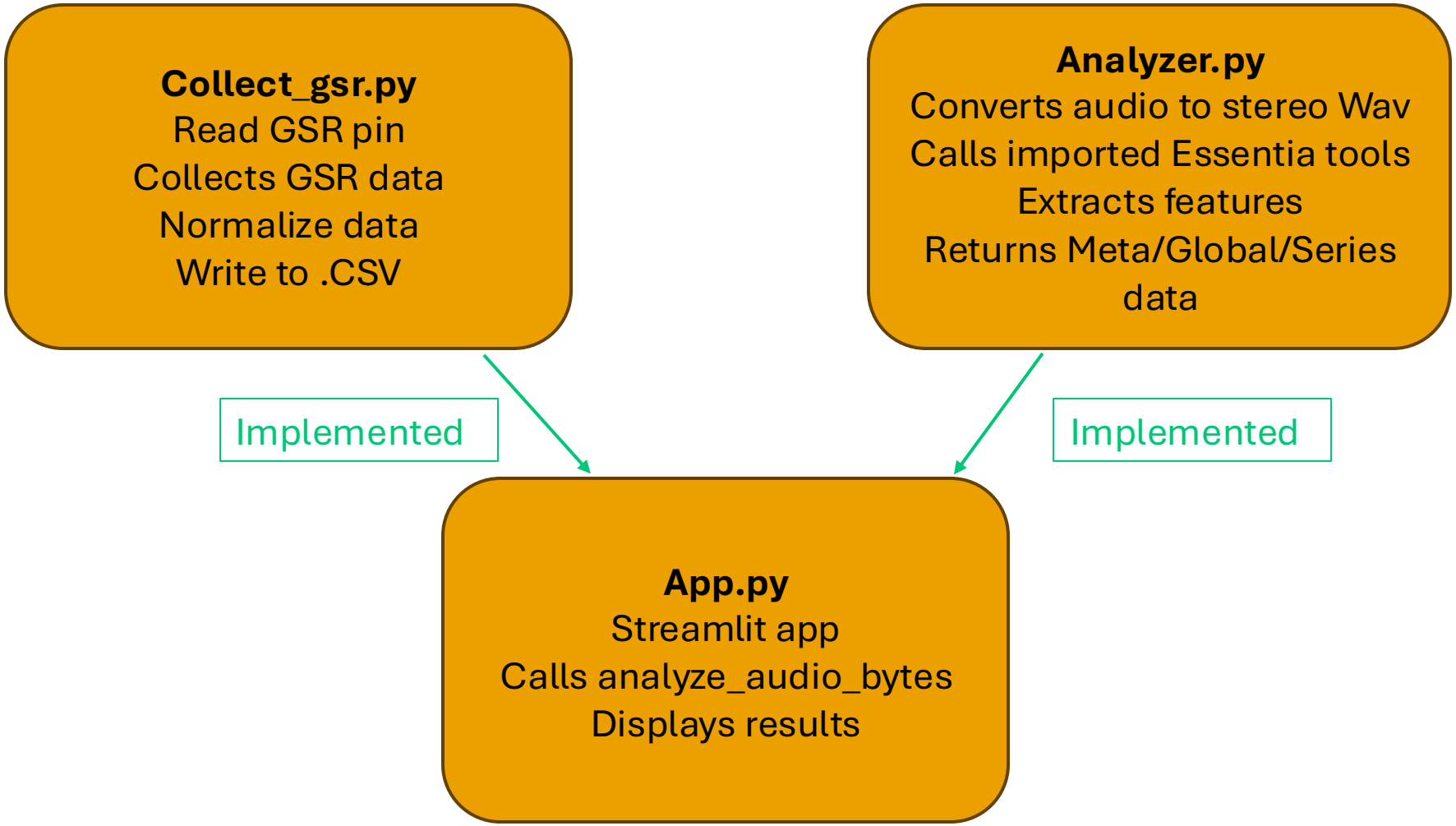
Smooth GSR



Song Analyzer + GSR (Pre-Integration)



Our new IMAGS Structure



Continuation of the Project

- Documented report:
  - Integrate MIR and Analysis to Existing IMAGS Framework
  - Add User Survey Questions
  - Expand Music Analysis
  - Explore More Sensors